Marmik Shah

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EXPERIENCE

Loylty Rewardz - A BillDesk Group Company *Apr 2025 – Present*
Senior Software Developer *Mumbai, India*

- Architected and implemented a new channel loyalty product for manufacturers and finance companies, serving DSAs, agents, and channel partners across the distribution lifecycle with KPI management, leaderboards, tier computation, customer segmentation, and an approval module.
- Engineered a real-time, API-driven microservices platform replacing T+1 bulk-file settlement, with 99% of transactions under 800ms; used Spring AOP for cross-cutting concerns and tracing, plus a webhook-based data-relay service.
- Built a low-latency balance service using caching, Bloom filters and database views, handling 300 TPS, letting partners fetch balances and burn points across corporates and LOBs in a single go.
- Developed a unified vendor integration layer for multiple modules using strategy-based orchestration and Kong gateway, cutting configuration and development effort by 60% while onboarding clients.
- Structured a bucket-based reward matrix framework and optimized enrichment and tenant loading, reducing database calls by over 45% and improving Hikari connection-pool use and startup time from 300s to 85s.
- Created AI-assisted admin panels and dev tools so client, product and RM teams manage configurations independently.
- Migrated services to Java 21 and handled SSL/TLS certificate migrations.

Software Developer *Jun 2023 – Mar 2025*

- Engineered a Kafka-driven ingestion platform processing 1M+ records daily, leveraging polling and webhook patterns for asynchronous status tracking and reconciliation to catch missed status updates.
- Designed an enterprise-wide loyalty program letting one program span multiple lines of business, with generic APIs.
- Developed a milestone-based journey rewards product on GraphQL; architected a profile-driven multi-database setup and a strategy-driven post-award layer for cashback and other redemption modules.
- Devised a parallel-processing lock for distributed ad hoc transactions and implemented the budget module to control reward spend, with threshold-based actions and real-time budget updates.
- Achieved zero downtime on production services with keep-alive, virtual IPs and master-slave HA; cut Redis memory by over 50% with a purging policy, a cluster setup and tuned RDB snapshots.
- Deployed microservices architecture on Kubernetes cluster; introduced observability using ELK, Prometheus, and Grafana.

ComED Learning *Apr 2022 – Jul 2022*
Software Development Intern *Remote*

- Built UI/UX components and server-side functionality for the ComED Learning platform using Django and ReactJS.
- Connected backend APIs to enable data exchange and dynamic content rendering; resolved application bugs and ran functional tests to make the platform more reliable.

KJ Somaiya College Of Engineering *Feb 2021 – Apr 2021*
Web Development Intern *Mumbai, India*

- Created the website for Symphony, the college cultural fest, collaborating with the cultural and design teams.
- Designed responsive web interfaces featuring interactive components and animations, tuning layouts to fit a range of screen sizes.

PROJECTS

Mumbai Police Portal — Created a full-stack complaint portal with station routing and status tracking (Django, PostgreSQL, Leaflet.js).

Waiterio Application — Implemented a client-server desktop app where waiters place orders and the chef fulfills (Java, Swing, Sockets).

Newzer — Developed a personalized news recommendation website that curates and summarizes news articles based on users' interests, preferences, and feedback using a recommendation algorithm. (Django, JavaScript, Machine Learning Algorithms).

EDUCATION

KJ Somaiya College Of Engineering, *B.Tech in Computer Engineering | CGPI – 8.93* 2021–2023
Mumbai University, *SSC – 89.80% | HSC – 84.62%* 2017–2019

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, C, C++, Solidity
Frameworks & Libraries: Spring, Spring Boot, Node.js, Django, React, Next.js, Vue.js, SASS, Bootstrap
Databases & Caches: MySQL, PostgreSQL, Greenplum, SQL Server, MongoDB, Supabase, Redis, Neo4j
AI Tools: Claude, GPT, Cursor, Kiro, Copilot
Devops & Infrastructure: , CI/CD, Jenkins, Prometheus, Grafana, New relic, ELK, Alertmanager, Linux, Bash, Nginx, Crontab
Tools & Platforms: Docker, Kubernetes, Git, Azure, Apache Camel, Airflow, Kong Gateway, IPFS, Parcel
Messaging & APIs: Apache Kafka, RabbitMQ, REST, JPA, GraphQL, JWT, OAuth2.0
Concepts: Microservices, Distributed Systems, Caching, High Availability, Bloom Filters, Agile, Blockchain, Multi Tenancy, UI/UX

ACHIEVEMENTS

Awards: Rising Star (2023–24), Extraordinary Diligence (2024–25), Think Out of the Box (2025–26).
Achievements: Won first place in HackerBash (CSI) hackathon as Team GrowMore. Project: Co-Vaccine Web portal (2021).